



First-Order Algorithms for Stochastic Multi-objective Optimization Problems

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Problem Formulation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a nonempty closed convex set. We study the **stochastic multi-objective optimization (SMOO)** problem:

$$\min_{\mathbf{x} \in \mathcal{X}} \left\{ F(\mathbf{x}) = (F_1(\mathbf{x}), \dots, F_m(\mathbf{x}))^\top \right\}. \quad (1)$$

For each $i \in [m] := \{1, \dots, m\}$, the function $F_i : \mathcal{X} \rightarrow \mathbb{R}$ is defined by $F_i(\mathbf{x}) := \mathbb{E}[f_i(\mathbf{x}, \xi)]$, where

- $\xi : \Omega \rightarrow \mathbb{R}^l$ denotes a random vector defined on a probability space $(\Omega, \mathcal{F}, P)$ with a probability measure P ;
- $f_i : \mathcal{X} \times \Xi \rightarrow \mathbb{R}$ is continuously differentiable in $\mathbf{x}$ for any fixed ξ with an $L_i(\xi)$ -Lipschitz continuous gradient ($\mathbb{E}[L_i(\xi)] < \infty$), and is measurable and integrable in ξ for any fixed $\mathbf{x}$, for each $i \in [m]$.

Sample Average Approximation (SAA) and Pareto Stationarity

SAA problem. Given i.i.d. samples $\{\xi_j\}_{j=1}^N$, for $i \in [m]$, we define $F_i^N(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N f_i(\mathbf{x}, \xi_j)$. We solve

$$\min_{\mathbf{x} \in \mathcal{X}} \left\{ F^N(\mathbf{x}) = (F_1^N(\mathbf{x}), \dots, F_m^N(\mathbf{x}))^\top \right\}. \quad (2)$$

Pareto optimality. $\mathbf{x}^* \in \mathcal{X}$ is called a Pareto optimal solution of problem (1) if there is no $\mathbf{x} \in \mathcal{X}$ such that

$$F(\mathbf{x}) \preceq F(\mathbf{x}^*) \quad \text{and} \quad F(\mathbf{x}) \neq F(\mathbf{x}^*).$$

The Pareto optimal solution of problem (2) can be defined similarly.

Pareto stationarity. Let $\Delta_m := \{w \in \mathbb{R}_+^m : w_1 + \dots + w_m = 1\}$.

$\mathbf{x}^* \in \mathcal{X}$ is called a Pareto stationary point of problem (1) if

$$0 \in \nabla F(\mathbf{x}^*) \Delta_m + \mathcal{N}_{\mathcal{X}}(\mathbf{x}^*).$$

$\mathbf{x}^* \in \mathcal{X}$ is called a Pareto stationary point of problem (2) if

$$0 \in \nabla F^N(\mathbf{x}^*) \Delta_m + \mathcal{N}_{\mathcal{X}}(\mathbf{x}^*).$$

- These are first-order necessary conditions for **Pareto optimality**.

Pareto-stationarity residual. For $\mathbf{x} \in \mathcal{X}$, define

$$\Theta(\mathbf{x}) := \text{dist}(0, \nabla F(\mathbf{x}) \Delta_m + \mathcal{N}_{\mathcal{X}}(\mathbf{x})).$$

Similarly, for the SAA problem, define

$$\Theta^N(\mathbf{x}) := \text{dist}(0, \nabla F^N(\mathbf{x}) \Delta_m + \mathcal{N}_{\mathcal{X}}(\mathbf{x})).$$

Projected Regularized Multi-gradient Step

A key step in our algorithm is to solve

$$\mathbf{p}_{\rho, \sigma}(\mathbf{x}; H) := \arg \min_{\mathbf{p} \in \mathcal{X} - \mathbf{x}} \left\{ \frac{\sigma}{2} \|\mathbf{p}\|^2 + \max_{\lambda \in \Delta_m} \left[\langle H\lambda, \mathbf{p} \rangle - \frac{\rho}{2\sigma} \left(\|\lambda\|^2 - \frac{1}{m} \right) \right] \right\},$$

for $H \in \mathbb{R}^{n \times m}$, $\rho > 0$, and $\sigma > 0$.

- $\mathbf{p}_{\rho, \sigma}(\mathbf{x}; H)$ is well-defined and unique.

Projected SAA-Regularized Multi-Gradient Descent Algorithm (SAA-RMGDA)

Require: $\mathbf{x}_0 \in \mathcal{X}$; $\sigma_0 > 0$; $\{\rho_k\}_{k \geq 0} \subseteq \mathbb{R}_{++}$. Set $k := 0$.

while stopping criterion not met **do**

(SAA gradients) For each $i \in [m]$, set $g_i^N(\mathbf{x}_k) := \frac{1}{N} \sum_{j=1}^N \nabla f_i(\mathbf{x}_k, \xi_j)$ and $G_k := [g_1^N(\mathbf{x}_k), \dots, g_m^N(\mathbf{x}_k)]$.

(RMGDA step) Compute $\mathbf{p}_k := \mathbf{p}_{\rho_k, \sigma_k}(\mathbf{x}_k; G_k)$ and $\mathbf{x}_{k+1} := \mathbf{x}_k + \mathbf{p}_k$.

(Regularization update) Set $\sigma_{k+1} := \sigma_k + \sigma_k \|\mathbf{p}_k\|^2$ and $k := k + 1$.

end while

Key features: function-value-free [1], line-search-free.

References

- [1] S. Gratton, S. Jerad, and P. Toint, "A stochastic objective-function-free adaptive regularization method with optimal complexity," *Open J. Math. Optim.*, vol. 6, pp. 1–24, 2025.
- [2] S. Liu and L. N. Vicente, "The stochastic multi-gradient algorithm for multi-objective optimization and its application to supervised machine learning," *Ann. Oper. Res.*, vol. 339, pp. 1119–1148, 2024.

Theoretical Guarantees

Assume that there exists $w \in \text{int } \Delta_m$ such that the function $\Phi^N(\mathbf{x}) := \sum_{i=1}^m w_i F_i^N(\mathbf{x})$ is level bounded on $\mathcal{X}$.

- This condition ensures the existence of a Pareto optimal solution of the SAA problem (2).

Boundedness of iterates.

Theorem 1. There exists a compact set $\mathcal{K}_N \subseteq \mathcal{X}$ that contains the sequence $\{\mathbf{x}_k\}$ generated by SAA-RMGDA.

Pareto-stationarity residual.

Theorem 2. Let $\mathcal{R}_k^N := \sigma_k \|\mathbf{p}_k\|$. We choose $\bar{t}_{N,K} \in \arg \min_{0 \leq t \leq K} \mathcal{R}_t^N$ and $\bar{\mathbf{x}}_{N,K} := \mathbf{x}_{\bar{t}_{N,K}+1}$. Then

$$\Theta^N(\bar{\mathbf{x}}_{N,K}) \leq \left(1 + \frac{L_G^N}{\sigma_0} \right) \min_{0 \leq t \leq K} \mathcal{R}_t^N,$$

and

$$\Theta(\bar{\mathbf{x}}_{N,K}) \leq \left(1 + \frac{L_G^N}{\sigma_0} \right) \min_{0 \leq t \leq K} \mathcal{R}_t^N + \sup_{\mathbf{x} \in \mathcal{K}_N} |\Theta^N(\mathbf{x}) - \Theta(\mathbf{x})|,$$

where L_G^N is the Lipschitz constant of ∇F^N on $\mathcal{K}_N$.

Convergence of SAA-RMGDA for SAA problem.

Theorem 3. For all $K \geq 1$, we have

$$\min_{0 \leq t \leq K} \mathcal{R}_t^N \leq O \left(\frac{1}{\sqrt{K+1}} \sum_{t=0}^K \rho_t \right).$$

- SAA-RMGDA converges to a Pareto stationary point of the SAA problem (2) whenever $\sum_{t=0}^K \rho_t = o(\sqrt{K})$.

Convergence of SAA on a compact set.

Theorem 4. Suppose, in addition, that $\mathcal{X}$ is compact. Let $\bar{\mathbf{x}}_N^* \in \mathcal{X}$ be an accumulation point of $\{\bar{\mathbf{x}}_{N,K}\}_{K \geq 1}$ for fixed N . Then any accumulation point of $\{\bar{\mathbf{x}}_N^*\}_{N \geq 1}$ is a Pareto stationary point of problem (1) with probability 1.

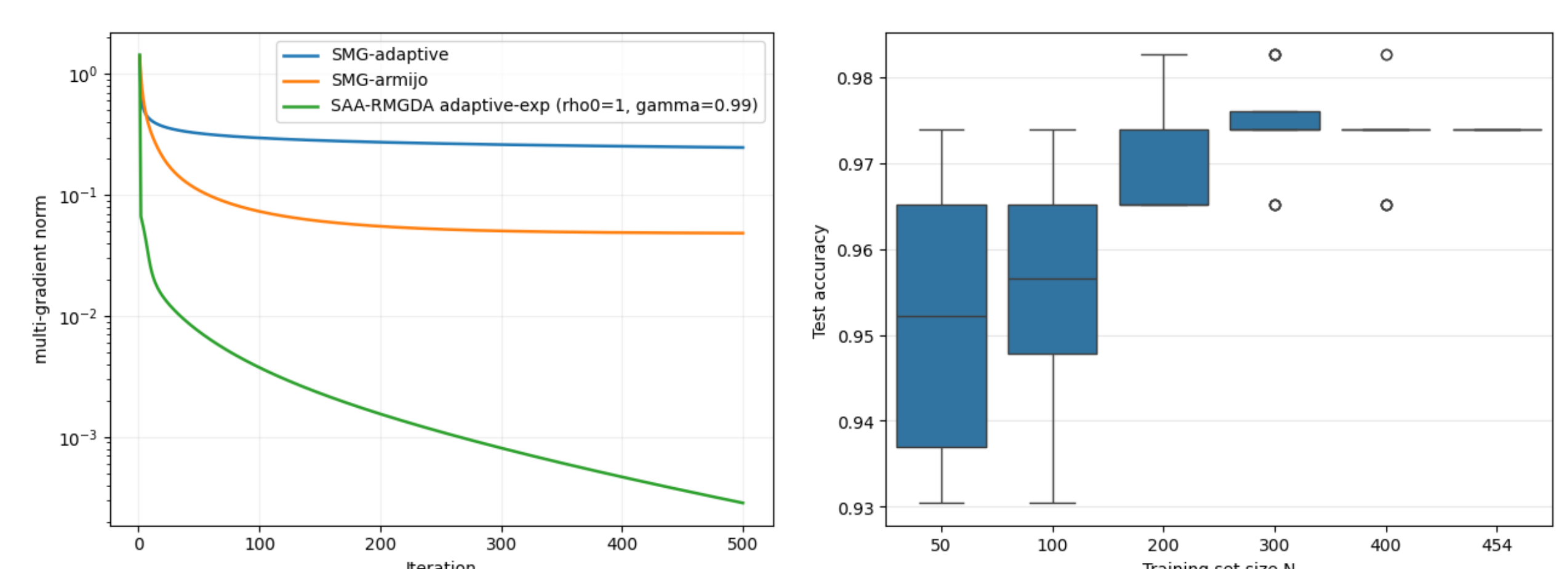
Numerical Experiment: Biobjective Logistic Regression

- **Dataset:** Wisconsin Breast Cancer for binary classification, with group label $g \in \{1, 2\}$, $n = 32$, and $m = 2$.
- **Model:** Linear classifier $h(\mathbf{x}; a) = z^\top a + b$, where $\mathbf{x} = (z^\top, b)^\top$.
- **SAA objectives:** Given samples $\{\xi_j = (a_j, y_j, g_j)\}_{j=1}^N$,

$$F_i^N(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N \log(1 + \exp(-y_j h(\mathbf{x}; a_j))) \mathbf{1}_{\{j: g_j = i\}} + \frac{\lambda_i}{2} \|\mathbf{x}\|^2.$$

- **Methods in comparison:**

- **SMG-diminishing** (stochastic multi-gradient descent with diminishing stepsize) [2].
- **SMG-armijo** (stochastic multi-gradient descent with multi-armijo line-search) [2].
- **SAA-RMGDA** ($\sigma_0 = 1$, $\rho_k = \gamma^k$, $\gamma = 0.99$).



Left: Stationarity measure versus iteration. Right: Test accuracy versus sample size N .

- Our algorithm significantly outperforms algorithms in [2].
- The test accuracy consistently improves as the sample size increases.